SiDTBF: Merging Soft Information With Dynamic Threshold Bit Flipping LDPC Decoding for 3-D NAND Flash Memory

Yangyi Li, Meng Zhang, Wei Li, Tianwei Gui, Changsheng Xie, *Member, IEEE*, and Fei Wu, *Member, IEEE*

***Abstract*— Through stacking and multibit technology, 3-D flash memory enhances storage capacity and density; nevertheless, the reduction in noise margin results in an increase in raw bit error rate (RBER) and a decrease in data reliability. Low-density parity-check (LDPC) codes are widely used in flash memory for improving data reliability because of its strong error correction capability. In the early stages of 3-D flash memory use, the RBER is low, and hard decision decoding (e.g., bit flipping (BF) decoding) is generally invoked for error correction. Existing LDPC codes with dynamic threshold BF (DTBF) decoding algorithms cannot correct bit errors when the gradually increasing RBER exceeds its error correction threshold, resulting in an increase in decoding latency. To enhance error correction capability and reduce decoding latency, this article proposes SiDTBF: merging soft information with DTBF LDPC decoding for 3-D NAND flash memory. First, the read reference voltage of various interval lengths is applied in accordance with the threshold voltage distribution drift characteristics of the 3-D flash memory cell to get the decoding soft information of each bit. Second, the strong and weak bits are distinguished using the soft information. In contrast to weak bits, which are more likely to be erroneous, strong bits are more likely to be correct. Finally, all the strong and weak bits are input as initial values for bit-flip iterative decoding. Using the column weight of the parity-check matrix, the threshold for the number of flipped weak bits is determined in the first decoding iteration process. The portion of the weak bits that exceeds the threshold is flipped. In the ensuing iteration phase, the DTBF decoding algorithm is used. SiDTBF improves decoding error correction performance by fusing each bit's soft information with the DTBF algorithm during the decoding phase. Simulation results show that compared with the current DTBF, SiDTBF significantly improves BF decoding error correction capability and reduces decoding latency.**

***Index Terms*—3-D NAND flash, capability, dynamic threshold bit flipping (DTBF), error correction, low-density parity-check (LDPC).**

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Yangyi Li, Meng Zhang, Wei Li, and Tianwei Gui are with the School of Computer Science and Technology, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: zgmeng@hust.edu.cn).

Changsheng Xie and Fei Wu are with Wuhan National Laboratory for Optoelectronics, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: cs_xie@hust.edu.cn; wufei@hust.edu.cn).

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I. Introduction

NAND flash-based solid state drives (SSDs) are used in many aspects of our daily lives. It is gradually taking the place of hard disk drives (HDDs) as the primary storage medium for electronic products due to its small size, low-power consumption, quick speed, and high capacity [1]. The primary drawback in comparison to HDD is the increased cost of storing per bit. The primary strategy used by manufacturers to lower storage expenses is to enhance storage density. It is challenging to further enhance the density of silicon-based planar flash memory due to the limitations of the quantum tunneling effect. The capacity of flash memory has significantly increased with the advent of 3-D stacking technology, which employs multilayer stacking and multiple bits in each cell [2]. However, it is more prone to noise, and data reliability—including retention time and program/erase (P/E) cycles—has emerged as a crucial issue that requires attention [3], [4]. Low-density parity-check (LDPC) codes are used as a trendy error correction code (ECC) in flash memory to increase the reliability of data due to their near-Shannon limit error correcting capacity [5], [6].

As an LDPC hard decoding method, the bit-flipping (BF) algorithm performs well in terms of low complexity and high throughput. A lot of comparison is required to determine the flipping threshold and flipping bit for the gradient descent BF (GDBF [7]), probabilistic GDBF (PGDBF [8]), multibit flipping (MBF [9]), adaptive threshold BF (ATBF [10]) in order to improve the BF algorithm's error correcting ability. The threshold is fixed by the dynamic threshold BF (DTBF) method, which shortens the comparison process but has little effect on error correcting ability [11]. The aforementioned approach improves the threshold selection and comparison time during the iteration, but it does not fully account for the features of the flash memory channel's threshold voltage distribution and the assistance of soft information to enhance the decoding performance [12].

This article develops SiDTBF: merging soft information (i.e., log-likelihood ratio (LLR) information per bit) with DTBF LDPC decoding for 3-D NAND flash memory. In the decoding process, soft information is used to improve the error correction ability of the DTBF decoding algorithm and reduce the decoding delay. To obtain the decoding soft information for each bit, the read reference voltage of different interval lengths is first applied in line with the 3-D flash memory cell's threshold voltage distribution drift characteristics. Second, the soft information is used to identify the *strong* and *weak* parts. *Strong* bits are more likely to be accurate than *weak* bits, which are more likely to be incorrect. In the end, bit-flip iterative decoding takes all of the *strong* and *weak* bits as input as beginning values.

In the first decoding iteration, the threshold for the number of flipped *weak* bits is established using the column weight of the parity-check matrix. Flipping occurs for the portion of the *weak* bits that surpasses the threshold. The DTBF decoding approach is applied in the next iteration step. By combining the DTBF algorithm with the soft information of each bit during the decoding stage, SiDTBF enhances decoding error correction performance and reduces decoding latency.

The main contributions of this article are as follows.

1) The observation results motivate us to propose an LDPC decoding scheme with strong error-correcting ability. Moreover, the read reference voltage is applied to obtain the soft information per bit to identify *strong* and *weak* bits during BF decoding corresponding to the unreliable bit by using the drift rule of the voltage distribution of 3-D flash memory cell.
2) The soft information and the DTBF algorithm are skillfully combined to improve the error correction ability and reduce the decoding delay. In the decoding process, the influence of soft information on the error correction performance of the DTBF algorithm is fully considered.
3) Simulation results show that the proposed decoding algorithm significantly improves error correction performance.

The remainder is set up as follows. In Section II, the background of flash memory and LDPC codes is presented. Section III gives the motivation of this article. Section IV provides a detailed description of the proposed SiDTBF; Section V gives the simulation results and analysis, and Section VI presents the conclusions.

II. Background

This section begins with an overview of NAND flash memory and error characteristics. Then, the fundamental concepts of LDPC codes are explained.

A. 3-D NAND Flash Memory

A flash memory block is made up of source selection line (SSL), drain selection line (DSL), wordlines (WLs), and bitlines (BLs), as illustrated in Fig. 1. Each WL comprises a storage series of logical pages, including the least significant (LSB), center significant (CSB), and most significant (MSB) pages [13], [14]. The storage cell is represented by the transistor at each WL and BL intersection [15], [16]. The smallest read/write data unit in flash memory is called a page, whereas the smallest erase unit is called a block, which is made up of several pages [17]. A string is formed of SSL, DSL, and all of the middle WL on a BL. The string's cells are joined in series. A page register, a cache register, and several blocks make up the plane. To avoid data transmission delays, data input and output are sent via the cache. The smallest unit of flash memory that can carry out commands and report its

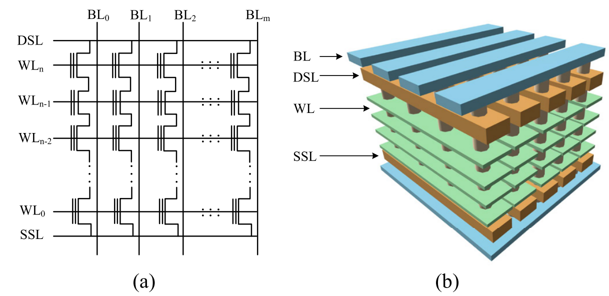


Fig. 1. Block structure of (a) 2-D and (b) 3-D NAND flash memory.

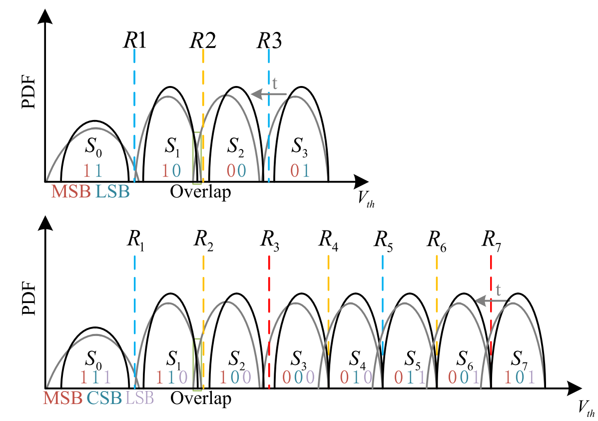


Fig. 2. Threshold voltage shift characteristics after programming and retention interference.

own status is called a logical unit (LUN), which is made up of many planes [18]. Flash memory's smallest storage unit is called a cell, and the number of charges in each cell serves as a representation of the various states of the current storage sequence [19].

Using multilevel cell (MLC) NAND flash memory as an example, each cell stores two bits of data, which are quantized into four logic states (i.e., "00," "01," "10," and "11") based on the amount of charge [20]. This is depicted in Fig. 2. A cell may do three fundamental operations: read, program, and erase. The program involves adding charge to the cell, erasure entails taking charge out of the cell, and read determines the current bit's state using the read reference voltage R [21], [22].

B. Channel and Interference in NAND Flash Memory

P/E cycles, read, layer fluctuation, and data retention are among the types of interference that can occur in 3-D NAND flash memory [4], [23].

Weak programming interference causes the threshold voltage distribution in the other layers to shift to the right when the current layer is read, which results in reading interference [24], [25]. Similar to reading interference, programming interference also weakly interferes with the current layer's unprogrammed cell in addition to affecting the other layers [26]. The influence of 3-D flash technology causes layer fluctuation, and the various physical attributes of distinct layers cause performance variations amongst them. Retention

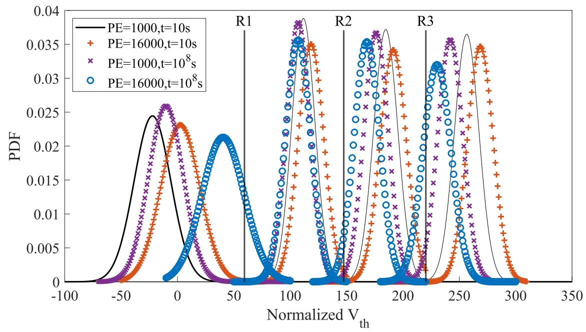


Fig. 3. NAND flash threshold voltage distribution under different P/E cycles and retention times.

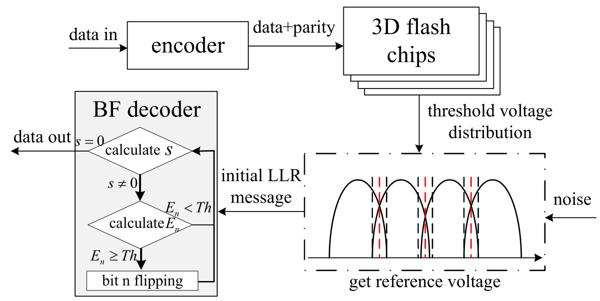


Fig. 4. Workflow of LDPC codes in 3-D NAND flash memory.

interference is caused by the charge tunneling effect, which causes the threshold voltage distribution of each cell to move to the left as the charge is lost as the charge retention in the flash memory increases over time.

Furthermore, the charge tunneling between the insulating layers during each P/E process wears down the tunnel oxide layer irreversibly, raising the threshold voltage distribution offset of different flash memory disturbances [27]. The threshold voltage distribution of MLC and TLC flash memory in each state without P/E and retention is depicted by the black lines in Fig. 2. Compared with MLC, TLC has a total of 8 states, each consisting of 3 bits. The threshold distribution of each state is more compact and is more likely to result in state overlap under the same P/E and retention times.

Following P/E cycles, nearby-state distributions overlap as retention time increases, as indicated by the gray lines. An error occurs when a cell in the overlapping region cannot accurately differentiate the storage state by using the read reference voltage. Long-term data retention further increases the risk of errors due to charge leaking. Charge loss causes the threshold voltage distribution to move to the left.

The quantity of charge in a transistor affects its charge leakage rate; the more charge, the faster the leakage rate, increasing the raw bit error rate (RBER) curvature on separate pages, with a maximum difference of more than ten times [28].

As seen in Fig. 3, the 3-D MLC NAND flash memory model presented by Luo et al. [29] is adopted as the channel model in this study. By using fitting functions, threshold voltage distributions under various P/E and retention times are identified.

The LLR information per bit is obtained by computing the logarithm value of the ratio between the probability of occurrence of the "1" bit and the probability of occurrence of the "0" bit in each group with the reference level acting as the boundary group, assuming P_i represents the probability distribution function of each cell state, where $i \in \{00, 01, 10, 11\}$. The left and right borders defined by the reference level are denoted by V_r and V_l, respectively, and the LLR values of the various pages in the MLC can be computed by exploiting formulas by the following equations [30]:

$$\mathrm{LLR}\,(\mathrm{MSB}) = \log \frac{\int_{\mathrm{Vr}}^{V_l} \left(P_{11}\,(x=1) + P_{10}\,(x=1)\right) \mathrm{d}x}{\int_{Vl}^{V_l} \left(P_{00}\,(x=0) + P_{01}\,(x=0)\right) \mathrm{d}x} \quad (1)$$

$$\mathrm{LLR}\,(\mathrm{LSB}) = \log \frac{\int_{\mathrm{Vr}}^{V_l} \left(P_{11}\,(x=1) + P_{01}\,(x=1)\right) \mathrm{d}x}{\int_{Vl}^{V_l} \left(P_{00}\,(x=0) + P_{10}\,(x=0)\right) \mathrm{d}x}. \quad (2)$$

Formulas (1) and (2) state that the correctness of the received data increases with the absolute value of LLR, and that the received data becomes less reliable when the closer the absolute value is near 0. Nine read reference voltage points can be applied to acquire the LLR information.

C. LDPC Coding

Three primary components make up the operation of LDPC codes in 3-D NAND flash memory: encoding, LLR information acquisition, and decoding [31]. As illustrated in Fig. 4, the data is encoded to add check bits, the initial LLR information is obtained based on the sampling level prior to decoding and iterative decoding is then carried out based on the initial LLR information. The decoding is successful when the syndrome s equals zero [5].

An $M * N$ dimensional parity-check matrix $\mathbf{H}$ can be used to define a binary regular (N, K) LDPC code [19]. The elements in the m-row and n-column of $\mathbf{H}$ are represented as $h_{m,n}$, where the number of rows $M \geq N - K$, N, and M indicate the length of the sequence after encoding and the syndrom sequence, respectively. The row (column) weight of each row (column) in $\mathbf{H}$ is a constant integer "1," which is stated as $d_v(d_c)$. The set of mth row "1" is represented as $A(n) = \{m, h_{m,n} = 1\}$. The estimated sequence is $y = (y_0, y_1, \ldots, y_{n-1})$ assuming the sequence $z = (z_0, z_1, \ldots, z_{n-1})$ is received through the flash channel.

Calculate the syndrome $s = z \cdot H^{\mathrm{T}} = (s_0, s_1, \ldots, s_{m-1})$ in order to determine if the received sequence fulfills the parity-check condition before applying the BF decoding algorithm. If not, calculate the flip function E_n corresponding to each bit based on the syndrome result.

The term E_n quantifies the reliability of each bit in the received codeword. For the parity-check matrix H, each row corresponds to a syndrome component s_i, while each bit aligns with a column in H. The reliability criterion is derived from the sum of s_i values from rows where the column entry is "1," and whether the bit has been flipped, which helps prevent consecutive erroneous flips of the same bit. In formula (3), α (defaulting to 2 in DTBF) acts as a weighting factor. The term $2s_m - 1$ maps the syndrome result to $+1, -1$, where a smaller E_n indicates higher reliability, and a larger E_n suggests lower reliability [11]

$$E_n = \alpha\,(z_n \oplus y_n) + \sum_{m\in A(n)} (2s_m - 1). \tag{3}$$

The fundamental principle of the BF algorithm operates on the basis that for a correctly transmitted codeword, all syndrome equations derived from the parity-check matrix must be satisfied. Given that the parity-check matrix is sparse with linearly independent rows, each bit affects the syndrome calculation results for the rows where its corresponding column contains "1." Since each bit participates in multiple parity-check equations, an erroneous bit will cause several syndrome equations to fail. The reliability of each bit is evaluated by examining its associated syndrome results. We quantify this reliability through E_n, which counts the number of unsatisfied syndrome equations (nonzero syndromes) for each bit position. Error correction is then performed by flipping bits whose E_n exceeds a predetermined threshold. Importantly, the value of E_n serves as an error probability indicator—the higher the E_n value, the greater the likelihood that the corresponding bit is erroneous.

In the BF algorithm, each iteration flips the corresponding bit positions of all the largest flipping functions, which has a relatively high probability of causing false flips. The DTBF algorithm adds a flipping threshold Th on the judgment of flipping bits to lower the possibility of error flipping and facilitate decoding convergence. Among them, initialize $Th = d_v + \alpha$, set t to the number of times $E_n \geq T_h$ occurs during the iteration. Within the error rate range that allows for complete decoding, as the number of iterations increases, the number of error bits decreases and the value of E_n decreases. To prevent the Th value from being too high, when there are errors, there are no bits that can be flipped. Throughout the iteration process, t is used to determine whether the Th value needs to be reduced. If $t = 0$, it indicates that there is no E_n higher than Th in this iteration, but there are error bits. Therefore, the Th value needs to be reduced before proceeding to the next iteration. First, a maximum Th value is set. As can be seen from the formula (3), the maximum is the sum of the column weight d_v of the parity matrix plus the weight factor α. The reduction step size in subsequent iterations needs to be obtained through actual testing to achieve the optimal result. $L0$ and $L1$ are used to determine the number of times Th is selected. To save the judgment logic, three Th values are generally set. When Th is iterated for $L0$ times, the Δ size is reduced, and after $L1$ iterations, the Δ size is further reduced. Generally, $L0 = 1$ and $L1 = 2$ are set as the default values.

III. Motivation

We test the actual 3-D TLC NAND flash memory chip with a floating gate structure. Table I displays the particular chip parameters. The primary control software, test board, high-temperature box, and flash memory chip are all part of the entire test platform. In order to carry out the experiment, the PC main control software transmits instructions to the test board via the SATA interface for erasing, programming, and reading the designated block or WL. The test board then connects to the flash chip to carry out the instructions and carry out the erasing and reading experiment.

TABLE I
3-D TLC NAND Flash Chip Information

Item	Configuration
Chip size	64GB
Logical unit(LUN)/Chip	2
Plane/LUN	4
Block/Plane	247
The number of layers	160
The number of wordline per Layer	10
Page size	16KB+2048B

First, generated random test patterns serve as the original encoded data. Flash memory programmed/read operations produce channel output. Comparisons are performed under identical P/E cycles and retention conditions. Second, compute bitwise differences between multiple channel outputs and original data, average the error counts across test iterations, and normalize by the codeword length to obtain RBER. Unlike Gaussian channels, flash memory exhibits distinct noise patterns. The proposed method captures the intrinsic error characteristics through empirical measurement.

The findings of the retention experiment were acquired at 100 °C by high temperature baking in order to expedite the collecting of experimental data following the corresponding room temperature retention duration [32].

We examine how the error condition of flash memory changes under various P/E cycles and retention times using the test platform. We next examine how various decoding methods affect BER as RBER rises. Initially, it is noted that the production method has a significant impact on the RBER differential across layers under the same wear condition, as well as the RBER of different pages on the same WL. As a result, there is a significant variation in the quantity of decoding iterations and decoding duration among pages of various levels within a block. The goals of this study are to decrease the discrepancy in decoding times between layers, enhance the performance of LDPC hard decoding, and close the performance gap between soft and hard decoding. This article's motivation is explained in full below.

Fig. 5 illustrates how the RBER varies across layers and pages of 3-D TLC NAND flash memory under various retention and P/E durations. In addition, the BER condition following LDPC soft decoding and hard decoding under various RBERs is examined. The objective is to systematically evaluate RBER characteristics under varying P/E cycles and retention conditions. Different blocks within the same flash device are assigned distinct P/E cycle counts. High-temperature acceleration is uniformly applied during retention testing. All functional blocks (excluding bad blocks) exhibit consistent error characteristics. The testing approach enables efficient characterization across multiple P/E conditions, simultaneous evaluation of retention effects through accelerated testing, and elimination of interblock variability as a confounding factor. The testing procedure comprises three key steps: 1) application of designated P/E cycles to respective blocks; 2) conducting accelerated retention testing under controlled conditions; and

3) comprehensive RBER measurement and analysis. As shown in Fig. 5, our analysis reveals layer- and page-dependent RBER variations in 3-D TLC NAND flash, comparative performance under different retention durations, and LDPC decoding efficiency across various RBER conditions.

The RBER of various pages in each layer of the subsequent block, P/E 1K, 5K, and 1w times, retention for 1 week, 3 months, and 1 year, respectively, are obtained using the aforementioned procedures. According to the experimental results, the RBER of the various layers in the block increases in a similar trend as the P/E cycle and retention time increase. However, the RBER of the various layers can differ by up to 16 times due to the production process and other factors, and the different pages use balanced gray code; the LSB page, CSB page, and MSB page have two read voltages and three read voltages, respectively, meaning that the RBER of the MSB page is higher.

It is important to note that due to the significant variation among layers, layers with low RBER require a significantly smaller number of decoding iterations than layers with high RBER [33], [34]. In fact, the RBER of some layers surpasses the hard decoding error correction capability, necessitating a switch to a soft decoding algorithm in order to achieve full error correction. In order to extract soft information for decoding, soft decoding necessitates a higher read voltage than hard decoding. There is a significant variation in reading performance across layers of flash memory due to the positive correlation between the read delay and the number of read voltages [35]. It is evident that RBER significantly affects LDPC's decoding performance. The number of iterations needed for decoding progressively rises as RBER increases. It is essential to raise the read voltage and employ soft decoding to fix errors when RBER exceeds 5×10^{-3} [19], [36]. For TLC, as shown in Fig. 5, RBER has surpassed the error correction capacity of hard decision decoding when P/E 1w times and retention time beyond 3 months. The soft decoding algorithm has a higher decoding time and necessitates more read voltages than the hard decoding approach.

Shannon's limit theorem [30] establishes that error-free decoding requires a signal-to-noise ratio (SNR) exceeding a threshold determined by the code rate and corresponding RBER below a critical level (derived from SNR conversion). Rapid FER reduction occurs when RBER falls below the threshold. Sharp transition reflects the coding gain near capacity limits. Existing implementations operate below Shannon capacity. Significant potential remains for algorithmic improvements. Potential approaches to enhance LDPC performance include advanced decoding algorithm development, optimized parity-check matrix design, improved reliability estimation techniques, and enhanced iterative processing methods.

It is evident that one significant aspect influencing flash memory reading performance is the LDPC code's decoding performance. In particular, the decoding delay difference between layers increases for blocks with a specific retention duration and P/E number because the error difference between layers is large and the same hard decoding algorithm is unable to fully correct the error information.

The hard decision decoding is mainly limited by the following fact: 1) restricted to binary ("0"/"1") input information; 2) incapable of processing multibit reliability (soft) information; and 3) significant performance gap compared to soft-decision decoding. Ryan and Lin [37] demonstrates a code rate of 0.89 with hard-decision threshold: 4.3 dB (AWGN channel) and corresponding RBER: 9×10^{-3}. The experimental results show the DTBF algorithm corrects up to 7×10^{-3} RBER, the proposed SiDTBF achieves 9×10^{-3} RBER correction (see Section V for the detailed different *bound* comparison). Under infinite code length, hard-decision decoding typically shows 1.5 dB performance degradation versus soft-decision. The performance gap increases substantially with practical (finite) code lengths.

Consequently, this study proposes the SiDTBF method. Originally, the DTBF algorithm could only receive hard information for decoding. By default, the error situation of each bit was of equal probability. Through the calculation of adjoint expressions and E_n, unreliable bits were flipped for iterative decoding. In fact, through the threshold distribution of flash memory, it can be concluded that the error probabilities of the incoming bits are different. The closer to the read voltage, the higher the error probability. Classifies bits into $strong/weak$ categories based on their voltage margins, incorporates this reliability information as soft inputs, and applies differentiated flipping probabilities during initial decoding iterations. This proposed approach specifically targets bits near read voltage thresholds (which have higher error probabilities) for prioritized correction, thereby significantly boosting the overall decoding performance compared to conventional hard-decision DTBF.

IV. Proposed SiDTBF Scheme

In this section, the proposed SiDTBF algorithm is first presented, including read reference voltage selection and the work process.

A. Overview

The fact that conventional hard decision decoding is insufficient to fully correct errors under the same P/E and retention time becomes a serious problem as flash density increases. In order to increase the DTBF algorithm's capacity for error correction, we suggest combining soft information with DTBF (SiDTBF). In Fig. 6, the suggested architecture is displayed.

Reliability thresholds (E_n^*) are established based on voltage distributions. $Weak$ bits are evaluated against the E_n^* threshold. Bits with $E_n > E_n^*$ are identified as unreliable and flipped. The conventional DTBF algorithm is employed for the remaining iterations. This proposed approach maintains computational efficiency while improving correction capability. The proposed method achieves enhanced error correction for hard-decision decoding, reduced performance gap compared to soft-decision approaches, and maintains the implementation simplicity characteristic of hard-decision systems.

B. Read Reference Voltage Selection

There are four states S_0, S_1, S_2, and S_3 in MLC. As illustrated in Fig. 7, we define the *bound* parameter as the ratio

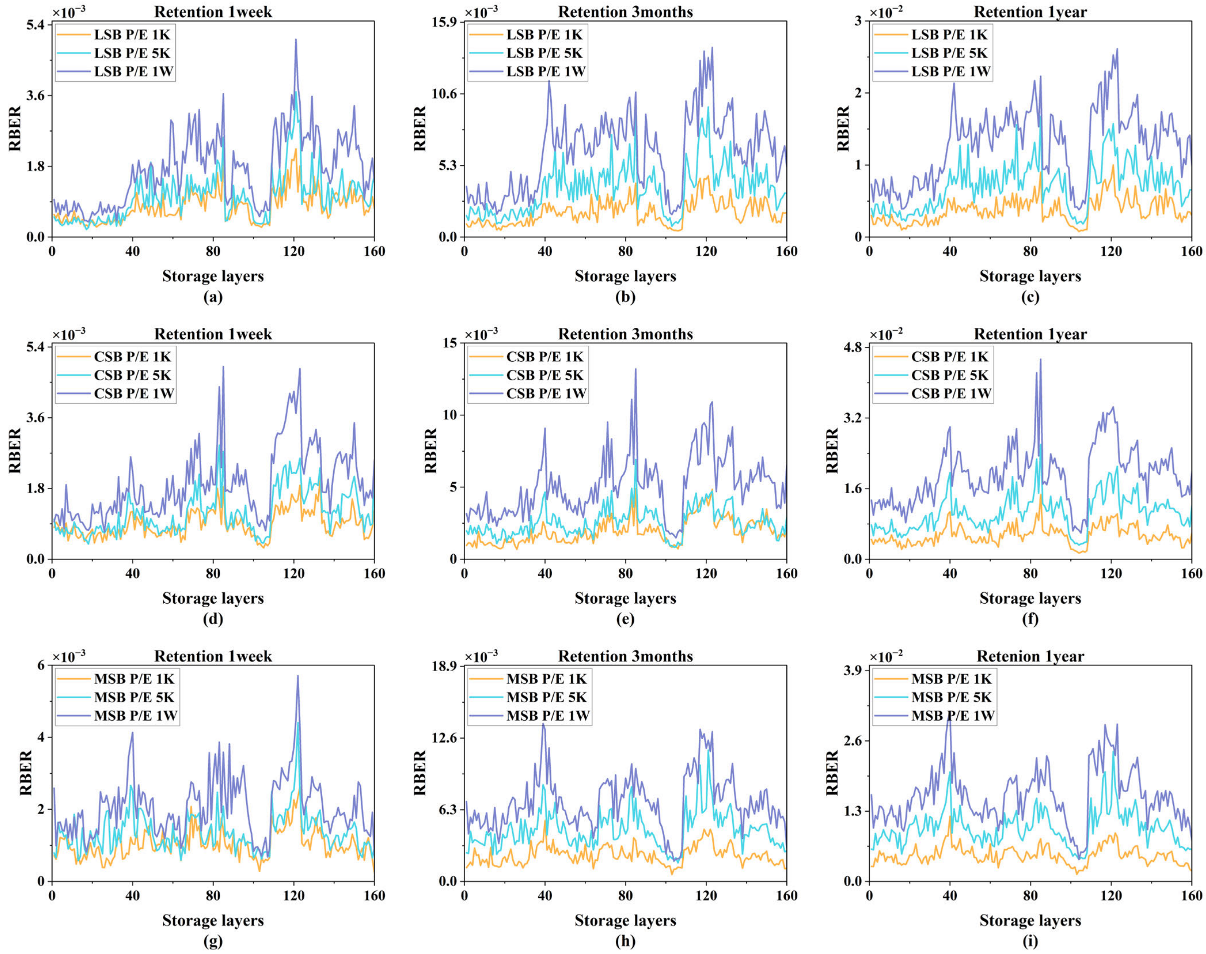


Fig. 5. Variations of RBER changes between all layers and pages in a block with different P/E cycles and retention times (i.e., (a)–(c) LP, (d)–(f) MP, and (g)–(i) CP).

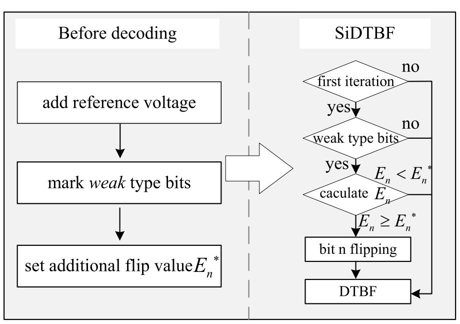


Fig. 6. SiDTBF algorithm workflow.

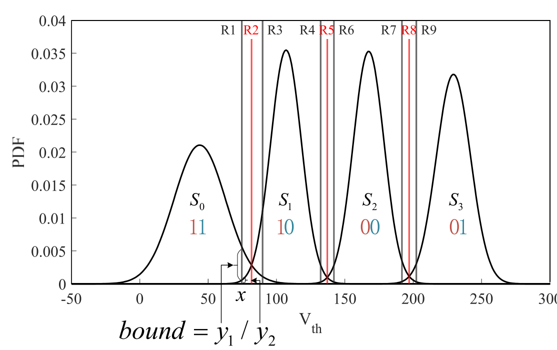


Fig. 7. 3-D MLC NAND flash threshold voltage distribution.

y_1/y_2 of probability density functions for adjacent states at voltage x within their overlapping region. This mathematically represents the likelihood ratio between competing states. The additional read voltage is optimized based on comprehensive 3-D NAND flash memory characterization. We employ two complementary approaches: 1) analytical modeling using the threshold voltage distribution formula from [29] Fig. 7; and 2) empirical characterization through actual chip testing under P/E cycling and retention conditions. Experimental validation compares error correction performance across different bound values. For empirical measurements, physical threshold distributions are obtained through chip scanning. Data is fit with Gaussian approximations for bound calculation. The optimal *bound* value is selected through systematic simulation testing. This dual approach combining theoretical modeling and experimental validation ensures robust voltage selection.

As shown in Fig. 7, in each overlapping area, the cell states between the two new read voltages can be considered unreliable, and the cells within this range are marked as *weak* type bits. On the basis of the original 2-bit state value, bit "0" is added to merge into a 3-bit number to indicate that the current cell value is unreliable. Cells outside this range are marked as *strong* type bits, and bit “1” is added to merge into a 3-bit number to represent that the current cell value is reliable. Taking Fig. 7 as an example, after adding the read voltage, the state values from left to right are “111,” “011,” “010,” “110,” “010,” “000,” “100,” “000,” “001,” and “101.”

The threshold voltage distribution of each state of flash memory is a Gaussian-like distribution. As P/E and retention increase, the states overlap near the reading voltage. The closer the state is to the reading voltage, the higher the error probability. Therefore, by adding new read voltages in the overlapping areas, the parts with higher error probabilities are distinguished, and the flipping probability of the unreliable parts is increased during decoding. The magnitude of the *bound* can reflect the overlap of the two states under the current voltage. The larger the *bound*, the farther it is from the hard information reading voltage, and the more overlapping areas are covered within the range. Therefore, raising the *bound* will increase the BF probability in the overlapping area, but this will also increase the number of correct bits covered. If the *bound* is too large, the number of correct bits within the range will be much higher than the number of incorrect bits. After flipping through E_n^*, the overall number of errors will increase. Therefore, it is necessary to select an appropriate *bound* value.

By adjusting the *bound* within the overlapping interval to select the position of the additional reference level, the data in the middle of the additional read voltages on both sides of each optimal read voltage is of the *weak* type bits, and vice versa is of the “strong” type bits. Increasing the flipping probability of the *weak* type bits can enhance the error correction ability of the DTBF algorithm.

Ultimately, the P/E cycle is modified to decrease 1000 times for each examination and obtain the reference level under different RBER, which is used to judge whether the same E_n^* can be used under different P/E cycles.

The data that reads the voltage in the extra read reference voltages interval on both sides of the optimal reference voltages is marked as unreliable data. The optimal reference level is utilized to derive the channel information for the DTBF algorithm, which is made of bits “0” and “1.” Using 3-D MLC NAND flash memory as an example, as illustrated in Fig. 7, data between reference voltages $R4$ and $R6$ is deemed unreliable for MSB pages and is designated as a *weak* type; similarly, data between reference voltages $R1$ and $R3$ and $R7$ and $R9$ are deemed unreliable for LSB pages and are designated as *weak* types. The *weak* types identifier sequence and the channel information are both sent to the decoder.

C. SiDTBF Work Process

Because the logic of the BF algorithm is to determine the reliability by calculating the number of nonzero adjoint expressions of each bit in the check matrix, flash memory can clearly show that the unreliability in the overlapping area is higher than that in other areas through the threshold voltage distribution. Therefore, reducing the E_n^* value can make the incorrect bits in the overlapping area easier to correct. E_n is set by default when the error probability of each bit is the same. E_n^* is set for regions with a higher error probability, so E_n will be higher than E_n^*. The E_n value has no direct relationship with the current bit error rate. Therefore, BF decoding will fail to complete the decoding even after reaching the maximum number of iterations due to inaccurate E_n values, which will cause the same bit to be flipped continuously during the iteration process. This is also the reason why the DTBF algorithm adds whether the bit has been flipped, that is, the $\alpha(z_n \oplus y_n)$ part, to the reliability judgment on the basis of the BF algorithm. Improved algorithms such as ATBF [10] and GDBF [7] for the BF algorithm are also optimizing the E_n value to prevent false flips from occurring. The formula of E_n^* is the same as that of E_n. Essentially, it provides a lower flipping threshold for unreliable regions, making the error bits in this region easier to correct in the first iteration. Therefore, the way to select the E_n^* value is shown in Fig. 7. Under the given P/E and retention conditions, traverse the value range of E_n^*, and select the E_n^* value that can correct the most bits without generating a large number of erroneous correction bits. Through this E_n^* value, it can be ensured that the number of erroneous bits in the overlapping area is reduced during the initial iteration process.

Then, we need to determine the optimal E_n^* value under different P/E and retention times. E_n^* is calculated using the formula: using formula $E_n^* \in \{-d_v - \alpha + k\Delta, k = 0, 1, 2, \ldots, \}$, where $\alpha = 2$ (empirically determined constant)

Algorithm 1 SiDTBF Decoding Process

Input: Soft input information for MSB and LSB $\mathbf{z}$
Output: indicates the decoding bit sequence $\mathbf{y}$
Set the maximum decoding iteration as $itermax$ and Initialize $iter = 1$, intermediate sequence $\mathbf{y} = \mathbf{z}$
for $iter = 1$ **to** $itermax$ **do**
  Calculate the adjoint result for each row in the check matrix $\mathbf{s} = \mathbf{y} \cdot H^{\mathrm{T}}$
  if $\mathbf{s} = \mathbf{0}$ **then**
    decoding success, record the number of iterations and output the decoding result $\mathbf{y}$
  else
    Calculate E_n for each bit, where $E_n = \alpha(z_n \oplus v_n) + \sum_{m \in A(n)} (2s_m - 1)$
    for i=1 **to** n **do**
      if decoding MSB page **then**
        if $iter = 1, b_{msbi} = 1$ and $E_i > E^*$ **then**
          $y_i = 1 - y_i$
        end if
      else
        if $iter = 1, b_{lsbi} = 1$ and $E_i > E^*$ **then**
          $y_i = 1 - y_i$
        end if
      end if
    end for
    Continue use the DTBF decoding algorithm to decode
  end if
end for
Output the result $\mathbf{y}$

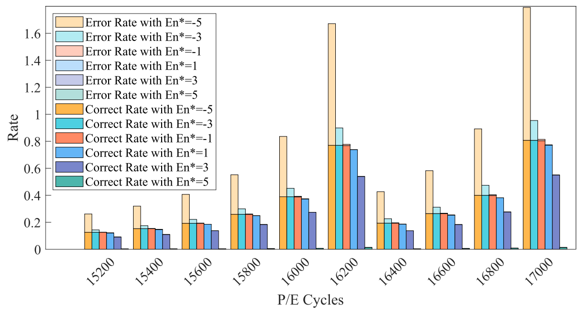


Fig. 8. Rate of the number of flipping errors and correct bits to the total number of errors under different P/E cycles and different E_n^*.

and $\Delta = 1$ (testing step size). We conduct comprehensive testing across various retention periods and P/E cycles. For each test condition, we evaluate error correction performance across multiple E_n^* values and select the optimal E_n^* that maximizes correctable errors while minimizing additional errors. There are several special considerations for retention effects. We perform detailed characterization at 10^8s retention, test P/E cycles ranging from 15 200 to 17 000, and analyze 10 000 frames per P/E cycle condition. Only the most effective E_n^* values for key retention points are stored in the decoder to balance complexity with performance by focusing on critical retention values. As shown in Fig. 8, this approach demonstrates P/E cycle independence in E_n^* selection and manages retention impact through selective optimization with consistent performance across operating conditions.

Because the errors of each state in the flash memory are concentrated in the overlapping area of the threshold voltage distribution, increasing the probability of *weak* bits flipping in the first iteration can reduce the number of errors in the overlapping area. Because the E_n value needs to be calculated, it cannot be completed before the decoding begins. After the first iteration, the error probability of the overlapping area decreases. Using the same method to increase the flipping probability again will lead to an increase in the probability of false flipping, which has no obvious effect on reducing the number of iterations and may even increase the number of iterations.

As illustrated in Fig. 8, the light-colored bars represent the ratio of flipped bits to actual errors among *weak* bits for different flipping thresholds (E_n^*). The dark-colored bars show the ratio of correctly corrected bits to actual errors. Optimal correction occurs when dark bars approach 1, indicating nearly all erroneous *weak* bits are properly corrected. The difference between light and dark bars represents the error flipping ratio under each E_n^*. Taller light bars indicate higher probabilities of erroneous flipping. This visualization enables us to evaluate the error correction performance for weak bits by analyzing both the correction efficiency and additional errors introduced at each threshold level.

In addition, it can be seen that under different P/E cycles, the proportion of additional errors that can be corrected and generated after adjusting the size of E_n^* is the same. P/E cycling induces structural wear in floating-gate transistors. Progressive degradation enhances charge tunneling susceptibility. Retention stress accelerates charge leakage through worn tunneling barriers. Charge loss during retention causes a leftward shift in Vth distributions. Identical materials and manufacturing processes ensure uniform wear characteristics. Experimental conditions maintain consistent stress across all layers. All layers exhibit similar RBER growth patterns under comparable stress. The correlation between P/E cycles, retention time, and RBER is consistently observed. Layer-to-layer variation remains minimal due to controlled experimental conditions. This systematic analysis demonstrates that the fundamental physical mechanisms governing charge retention lead to predictable RBER evolution in flash memory arrays.

We thus propose the SiDTBF decoding method. The main idea behind this method is to optimize the decision mechanism in the DTBF algorithm decoding process by introducing soft information.

The specific process is as described in Algorithm 1. Before adding the read voltage, the hard information "0" and "1" is input to the decoder, represented by the z sequence. After adding the read voltage, in addition to the original hard information, 1-bit information for judging the strength of the bit needs to be input. Taking MLC as an example, define two sequences $B_{\text{lsb}} = (b_{\text{lsb1}}, b_{\text{lsb2}}, \ldots, b_{\text{lsbn}})$ and $B_{\text{msb}} = (b_{\text{msb1}}, b_{\text{msb2}}, \ldots, b_{\text{msbn}})$. For MSB, the bit between $R4$ and $R6$ is judged as a *weak* type, set $b_{\text{msb1}} = 0$, and the rest are set to $b_{\text{msb1}} = 1$. For LSB, the bits between $R1$ and $R3$ and $R7$ and $R9$ are judged as a *weak* type, $b_{\text{lsbi}} = 0$, and the rest are set to $b_{\text{lsbi}} = 1$. During decoding of different pages, B_{lsb} and B_{msb} are input together with the z sequence to the decoder. An E_n^* is added inside the decoder to judge the *weak* bits. y represents the decoding result in each iteration. Set the initial information $y = z$, and enter the iteration. First, calculate the resultant value $s = y * H$. If $s = 0$, the decoding is successful and the decoding result y is output. If not, proceed to the next iteration. The flip function value E_n of each bit is calculated through formula (3). Different from DTBF, in the initial iteration process, the flip value E_n corresponding to the bit with $b_i = 1$ is compared with E_n^*, if $E_n > E_n^*$, the y_i value is flipped, and the other bits are still compared with the Th value. In subsequent iterations, the DTBF algorithm is used for decoding.

This optimized approach provides several advantages: 1) maintains the computational efficiency of DTBF; 2) improves initial error correction through intelligent bit classification; and 3) preserves the simplicity of the original algorithm while enhancing performance.

As shown in Fig. 8, setting a suitable E_n^* can guarantee that, under varying RBER, the number of errors removed is greater than the number of additional errors introduced, even though more error bits will be introduced. In addition, the number of iterations needed to finish the decoding process is decreased because there are fewer errors overall in the succeeding iteration phase. Thus, SiDTBF can enhance decoding performance and decrease the decoding iteration time.

V. Performance Evaluation

In this section, we describe and analyze the simulation results of the SiDTBF algorithm in terms of BER, FER, and

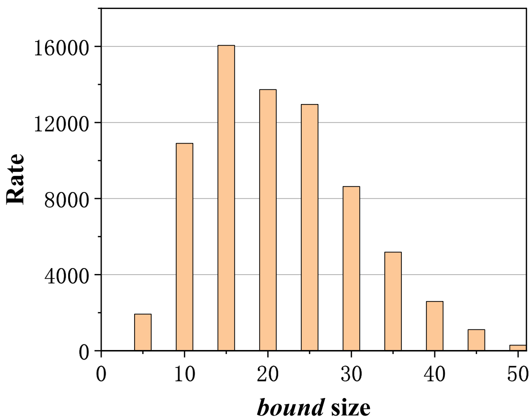


Fig. 9. SiDTBF uses the ratio of different *bound* and DTBF to BER when RBER is 0.08.

decoding delay. We use computer simulations to evaluate the error correction capabilities and decoding latency of SiDTBF.

First, we compare the DTBF and SiDTBF algorithms with the code length K of 32 768, code rate of 0.89, column weight d_v of 5, and $E_n{}^* = -d_v + 4$ [11]. Fig. 10 shows the performance of DTBF and SiDTBF for *bound* sizes of 5, 10, 15, 20, and 50. It is evident that adding *bound* improves error correction performance and reduces decoding latency compared to the DTBF algorithm.

In order to have a more intuitive understanding of the gain of the DTBF algorithm under the same E_n^* with different bound values, as shown in Fig. 9, the Y-axis is the ratio of the BER decoded by the SiDTBF algorithm with different bound values under the condition of RBER $= 8 \times 10^{-3}$ to the BER decoded by the DTBF algorithm. The higher the ratio, the higher the gain, and the lower the BER after decoding under the same RBER.

The results indicate that the optimal decoding performance can be obtained by setting *bound* to 20 in the current threshold voltage distribution model. Specifically, the FER performance increases from 6×10^{-1} to 4×10^{-5} at an RBER of 6.8×10^{-3}, and the decoding time decreases from 0.017 to 0.0053 s.

This is due to the fact that when the *bound* increases at the beginning, the position of the additional reference voltage is in the part of the overlapping region with the most concentrated error bit. The DTBF algorithm's characteristics state that the probability of the error bit's E_n in the region with the error concentration is greater than 50%, meaning that as the *bound* increases, there are more correct bits by E_n than error flipping bits. In the first iteration of the SiDTBF algorithm, there are more additional corrected errors and fewer error flipping bits than correct corrected bits. As the *bound* range is increased further, the error bit density falls and the likelihood that the error bits' flipping threshold will be higher than the proper bits falls in the range of extra reference voltage. At this point, raising the *bound* value will result in a likelihood of the error flipping that is greater than 50%, which will lower the ability to fix errors.

Thus, choosing the location of the extra reference voltage before the error bit proportion drops in the overlapping interval will result in the best decoding performance. The region with the most concentrated error bit can be identified and the likelihood of error flipping decreased by using simulation to choose the proper *bound* value.

Next, we evaluate the impact of E_n^* between -5 and 5 on decoding speed in the case of *bound* 20. Since we need to flip the bit in the overlap area that is higher than E_n^*, we must select the proper E_n^* value to avoid error flipping situations. To determine the change in the error correction ability of E_n^* between -5 and 5, we choose *bound* value 20 with the highest performance, as shown in Fig. 11.

As shown in Fig. 11, when $E_n{}^* = 5$, there is no additional bit flip, and the error correction ability is the same as the DTBF algorithm. With the decrease of E_n^*, the error correction ability gradually increases. When $E_n{}^* = -1$, the error correction ability is much stronger, it can be considered that when $E_n{}^* = -1$, the difference between the number of extra correct bits by using the SiDTBF algorithm and the number of error flipping bits is much larger. As E_n^* continues to decrease, the error correction capability of SiDTBF begins to decrease, and when $E_n{}^* = -5$, the error correction capability of SiDTBF is lower than that of DTBF. This is because the E_n computed by the correct bit is not precisely the minimal value $-d_v - \alpha$, and the E_n grows when it is influenced by the error bit. Therefore, reducing $E_n{}^*$ will increase the probability of error flipping.

Raising $E_n{}^*$, on the other hand, guarantees that there is no error flipping but also lowers the quantity of flipping error bits, which impacts the decoding result. Furthermore, it is discovered that when $E_n{}^* = -5$, the error correcting ability is even less than DTBF since additional overcorrection bits are generated through preprocessing. This conforms to the phenomenon in Fig 8, when E_n^* reaches the critical value, continuing to reduce E_n^* does not increase the number of error corrections but increases the probability of error flipping. Compared with DTBF, it can be found that the error correction ability is even lower than DTBF when $E_n^* = -5$, because the number of error flips is more than the number of correct flips. In addition, it can be confirmed from Fig. 8 that the optimal *bound* value is the same in different RBER cases, so for chips with the same structure, the optimal *bound* value obtained from the test can be applied to the decoding as a fixed value without dynamic adjustment.

It is evident that varying $E_n{}^*$ has a greater effect on error correction ability than *bound* because, when P/E cycles and retention periods are taken into account, read errors that arise during data reading are primarily focused in the overlapping region. There is little variation in the number of errors that can be covered after the *bound* is raised to an extent, and E_n^* is used to evaluate the error probability of the current bit. Adjusting the value of E_n^* will directly affect the accuracy of judging the error bits, and an inappropriate E_n^* value will lead to the failure to accurately judge the error bits and increase the error probability. Therefore, *bound* and E_n^* are two crucial parameters in the SiDTBF algorithm. SiDTBF improves decoding performance well beyond the DTBF algorithm without adding to the decoding complexity by merging with soft information.

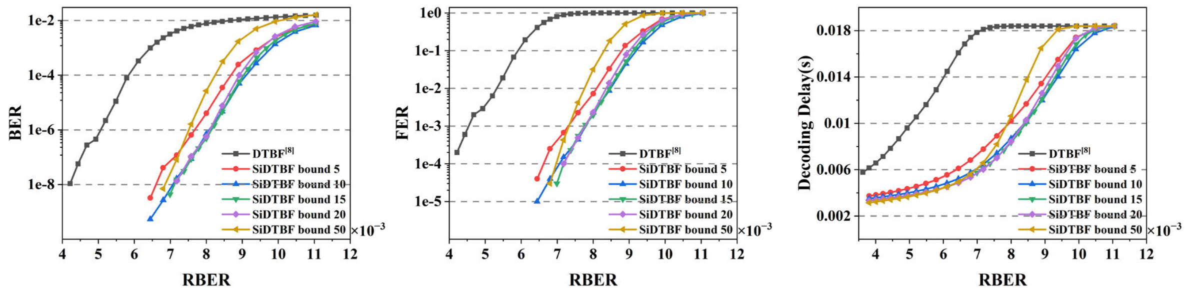


Fig. 10. Variations of BER, FER, and decoding delay of DTBF and SiDTBF algorithms with different *bound* under varying RBERs.

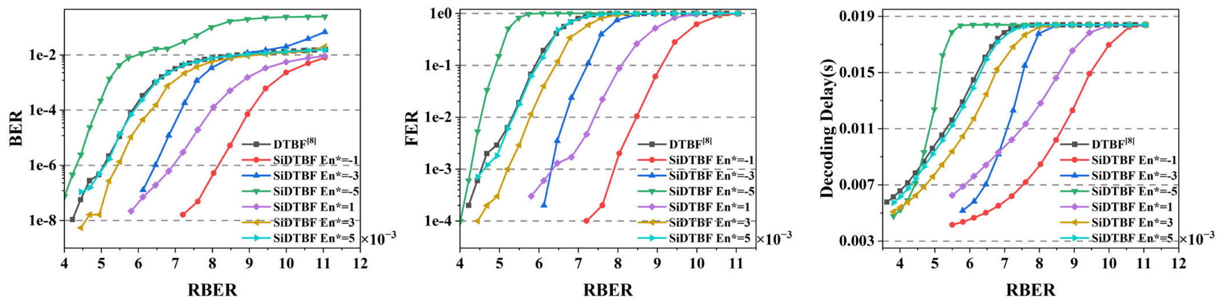


Fig. 11. Variations of BER, FER, and decoding delay of DTBF and SiDTBF algorithms with different $E_n{}^*$ under varying RBERs.

VI. Conclusion

This article proposes the SiDTBF scheme that effectively improves the hard decoding ability of the LDPC code. Combining with the characteristics of the flash memory channel, the decoding ability of the DTBF algorithm is effectively improved through soft information input. We initially examine how choosing various extra read reference levels affects SiDTBF decoding error correction capability. Then, by utilizing the error bit with a larger flipping function value in the overlapping interval, the SiDTBF method with soft information is presented in combination with the DTBF algorithm. By supplying more soft information to flip more error bits in the first iteration, SiDTBF can enhance the error correction performance. The results of the simulation demonstrate that the RBER starting point of SiDTBF can be corrected is increased from 7×10^{-3} to 1×10^{-2} compared to DTBF, and can reduce the number of iterations of decoding and the decoding delay.

References

[1] J. Cui, K. Tang, and L. T. Yang, "A fast secure deletion strategy for high-density flash memory through WOM-v codes," in *Proc. 60th ACM/IEEE Design Autom. Conf. (DAC)*, Jul. 2023, pp. 1–6.

[2] Q. Li et al., "Shaving retries with sentinels for fast read over high-density 3D flash," in *Proc. 53rd Annu. IEEE/ACM Int. Symp. Microarchitecture (MICRO)*, Oct. 2020, pp. 483–495.

[3] S. I. Shim, J. Jang, and J. Song, "Trends and future challenges of 3D NAND flash memory," in *Proc. IEEE Int. Memory Workshop (IMW)*, May 2023, pp. 1–4.

[4] S. K. Park and J. Moon, "Characterization of inter-cell interference in 3D NAND flash memory," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 68, no. 3, pp. 1183–1192, Mar. 2021.

[5] T.-Y. Wang, C.-W. Tsao, Y.-H. Chang, and T.-W. Kuo, "Retention-aware read acceleration strategy for LDPC-based NAND flash memory," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 42, no. 12, pp. 4597–4605, Dec. 2023.

[6] Y. Du, S. Huang, Y. Zhou, and Q. Li, "Towards LDPC read performance of 3D flash memories with layer-induced error characteristics," *ACM Trans. Design Autom. Electron. Syst.*, vol. 28, no. 3, pp. 1–25, May 2023.

[7] T. Wadayama, K. Nakamura, M. Yagita, Y. Funahashi, S. Usami, and I. Takumi, "Gradient descent bit flipping algorithms for decoding LDPC codes," *IEEE Trans. Commun.*, vol. 58, no. 6, pp. 1610–1614, Jun. 2010.

[8] O. A. Rasheed, P. Ivaniš, and B. Vasic, "Fault-tolerant probabilistic gradient-descent bit flipping decoder," *IEEE Commun. Lett.*, vol. 18, no. 9, pp. 1487–1490, Sep. 2014.

[9] J. Jung and I.-C. Park, "Multi-bit flipping decoding of LDPC codes for NAND storage systems," *IEEE Commun. Lett.*, vol. 21, no. 5, pp. 979–982, May 2017.

[10] J. Cho and W. Sung, "Adaptive threshold technique for bit-flipping decoding of low-density parity-check codes," *IEEE Commun. Lett.*, vol. 14, no. 9, pp. 857–859, Sep. 2010.

[11] M. Jiang and D. Fan, "A low-latency BF decoding of LDPC codes with dynamic thresholds," *IEEE Commun. Lett.*, vol. 25, no. 9, pp. 2781–2785, Sep. 2021.

[12] Y. Li, G. Han, S. Huang, C. Liu, M. Zhang, and F. Wu, "Exploiting metadata to estimate read reference voltage for 3-D NAND flash memory," *IEEE Trans. Consum. Electron.*, vol. 69, no. 1, pp. 9–17, Feb. 2023.

[13] M. Zhang et al., "BeLDPC: Bit errors aware adaptive rate LDPC codes for 3D TLC NAND flash memory," in *Proc. Design, Autom. Test Eur. Conf. Exhib. (DATE)*, Mar. 2020, pp. 302–305.

[14] W. Liu et al., "Modeling of threshold voltage distribution in 3D NAND flash memory," in *Proc. Design, Autom. Test Eur. Conf. Exhib. (DATE)*, Feb. 2021, pp. 1729–1732.

[15] K. Hu, H. Hu, P. Han, and G. Han, "MCBGC: A multi-threshold copyback-based garbage collection scheme for 3D NAND flash memory," in *Proc. Int. Conf. Netw., Archit. Storage (NAS)*, Nov. 2024, pp. 1–4.

[16] Q. Hai, S. Zhang, C. Liu, and G. Han, "Hard disk drive failure prediction based on GRU neural network," in *Proc. IEEE/CIC Int. Conf. Commun. China (ICCC)*, Aug. 2022, pp. 696–701.

[17] Y. Wang, D. Wei, M. Liu, H. Feng, and L. Qiao, "EBDN: Entropy-based double nonuniform sensing algorithm for LDPC decoding in TLC NAND flash memory," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 43, no. 6, pp. 1914–1918, Jun. 2024.

[18] R. Micheloni, *3D Flash Memories*. Cham, Switzerland: Springer, 2016.

[19] M. Zhang et al., "ALCod: Adaptive LDPC coding for 3-D NAND flash memory using inter-layer RBER variation," *IEEE Trans. Consum. Electron.*, vol. 69, no. 4, pp. 1068–1081, Apr. 2023.

[20] S. Aritome, *NAND Flash Memory Technologies*. Hoboken, NJ, USA: Wiley, 2015.

[21] Y. Lv et al., "MGC: Multiple-gray-code for 3D NAND flash based high-density SSDs," in *Proc. IEEE Int. Symp. High-Perform. Comput. Archit. (HPCA)*, Feb. 2023, pp. 122–136.

[22] Q. Li et al., “Midas touch: Invalid-data assisted reliability and performance boost for 3D high-density flash,” in *Proc. IEEE Int. Symp. High-Perform. Comput. Archit. (HPCA)*, Mar. 2024, pp. 657–670.

[23] Y. Cai, E. F. Haratsch, O. Mutlu, and K. Mai, “Error patterns in MLC NAND flash memory: Measurement, characterization, and analysis,” in *Proc. Design, Autom. Test Eur. Conf. Exhib. (DATE)*, Mar. 2012, pp. 521–526.

[24] H. Feng, D. Wei, Y. Wang, Y. Song, Z. Piao, and L. Qiao, “Lightweight read reference voltage calibration strategy for improving 3-D TLC NAND flash memory reliability,” *IEEE Trans. Device Mater. Rel.*, vol. 23, no. 3, pp. 370–379, Sep. 2023.

[25] D. Wei et al., “Edge word-line reliability problem in 3-D NAND flash memory: Observations, analysis, and solutions,” *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 31, no. 6, pp. 861–873, Jun. 2023.

[26] J. Cui, F. Chen, L. Li, S. Nie, and L. T. Yang, “SmartNetSSD: Exploiting path resources for read performance improvement in network-based SSDs,” in *Proc. IEEE 42nd Int. Conf. Comput. Design (ICCD)*, Nov. 2024, pp. 356–359.

[27] B. Gu, L. Luo, Y. Lv, C. Li, and L. Shi, “Dynamic file cache optimization for hybrid SSDs with high-density and low-cost flash memory,” in *Proc. IEEE 39th Int. Conf. Comput. Design (ICCD)*, Oct. 2021, pp. 170–173.

[28] Y. Cai, Y. Luo, E. F. Haratsch, K. Mai, and O. Mutlu, “Data retention in MLC NAND flash memory: Characterization, optimization, and recovery,” in *Proc. IEEE 21st Int. Symp. High Perform. Comput. Archit. (HPCA)*, Feb. 2015, pp. 551–563.

[29] Y. Luo, S. Ghose, Y. Cai, E. F. Haratsch, and O. Mutlu, “Improving 3D NAND flash memory lifetime by tolerating early retention loss and process variation,” *ACM SIGMETRICS Perform. Eval. Rev.*, vol. 46, no. 1, pp. 1–48, Jan. 2019.

[30] P. Sweeney, *Error Control Coding*. Upper Saddle River, NJ, USA: Prentice-Hall, 1991.

[31] W. Zhao, H. Sun, M. Lv, G. Dong, N. Zheng, and T. Zhang, “Improving min-sum LDPC decoding throughput by exploiting intra-cell bit error characteristic in MLC NAND flash memory,” in *Proc. 30th Symp. Mass Storage Syst. Technol. (MSST)*, Jun. 2014, pp. 1–6.

[32] M. Ye, Q. Li, D. Wen, T.-W. Kuo, and C. J. Xue, “WearMeter: An accurate wear metric for NAND flash memory,” in *Proc. 29th Asia South Pacific Design Autom. Conf. (ASP-DAC)*, Jan. 2024, pp. 442–447.

[33] Q. Li, L. Shi, Y. Cui, and C. J. Xue, “Exploiting asymmetric errors for LDPC decoding optimization on 3D NAND flash memory,” *IEEE Trans. Comput.*, vol. 69, no. 4, pp. 475–488, Apr. 2020.

[34] B. Bao, Q. Li, W. Guan, Q. Wang, L. Liang, and X. Qiu, “Adaptive granularity progressive LDPC decoding for NAND flash memory,” *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 43, no. 4, pp. 1312–1316, Apr. 2024.

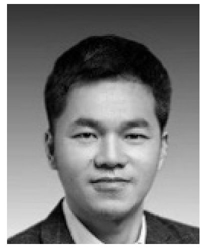
[35] Y. Du, D. Zou, L. Qiao, S. Liang, and C. J. Xue, “LaLDPC: Latency-aware LDPC for read performance improvement of solid state drives,” in *Proc. 33rd Int. Conf. Massive Storage Syst. Technol. (MSST)*, 2017, pp. 1–11.

[36] B. Chen, Y. Kong, Z. Sun, X. Fang, X. Zhan, and J. Chen, “High-to-Low flippling (HLF) coding strategy in Triple-levell-cell (TLC) 3D NAND flash memory to construct reliable image storages,” in *Proc. 6th IEEE Electron Devices Technol. Manuf. Conf. (EDTM)*, Mar. 2022, pp. 336–338.

[37] W. Ryan and S. Lin, *Channel Codes: Classical and Modern*. Cambridge, U.K.: Cambridge Univ. Press, 2009.

Yangyi Li is currently pursuing the Ph.D. degree with the School of Computer Science and Technology, Huazhong University of Science and Technology (HUST), Wuhan, China.

His current research interests include flash memory reliability and LDPC codes.



Meng Zhang received the Ph.D. degree in computer science and technology from the Huazhong University of Science and Technology (HUST), Wuhan, China, in 2019.

He is currently an Associate Professor with the School of Computer Science and Technology, HUST. His current research interests include nonvolatile memory (NVM), dynamic random access memory (DRAM), optical storage, and error correction codes.

Wei Li is currently pursuing the M.S. degree with the School of Computer Science and Technology, Huazhong University of Science and Technology (HUST), Wuhan, China.

His current research interests include flash memory reliability, dynamic random access memory (DRAM), and error correction codes.

Tianwei Gui is currently pursuing the Ph.D. degree with the School of Computer Science and Technology, Huazhong University of Science and Technology (HUST), Wuhan, China.

His current research interests include Reed–Solomon codes and optical storage reliability.

Changsheng Xie (Member, IEEE) received the B.S. and M.S. degrees in computer science and technology from Huazhong University of Science and Technology (HUST), Wuhan, China, in 1982 and 1988, respectively.

He is currently a Professor with Wuhan National Laboratory for Optoelectronics, HUST. His research interests include new storage technology and architecture, multimedia computing and networks, computer storage, and network storage and security.

Dr. Xie was the Deputy Director of Wuhan National Laboratory for Optoelectronics, the Deputy Director of Computer Peripheral Equipment Committee of China Computer Federation, a Committee Member of Information Storage of China Computer Society.

Fei Wu (Member, IEEE) received the B.S. and M.S. degrees in electrical automation from Wuhan University of Technology, Wuhan, China, in 1997 and 2000, respectively, and the Ph.D. degree in computer science from the Huazhong University of Science and Technology (HUST), Wuhan, in 2005.

She is currently a Professor with the Information Storage Laboratory, Wuhan National Laboratory for Optoelectronics, HUST. Her research interests include computer architecture, nonvolatile memory (NVM), embedded systems, and intelligent storage.